

Advanced Probabilistic Machine Learning

Approximate Inference: Variational Inference

Ralf Herbrich, Rainer Schlosser

Overview

1. Minimizing Reverse KL-Divergence via ELBO
2. Variational Inference for Gaussian Factors
3. Application: Robust Bayesian Regression and Ranking

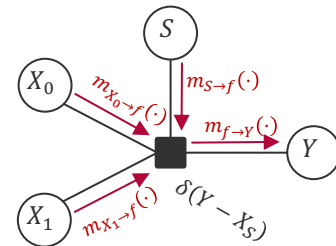
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*Unit 6 – Approximate Inference:
Variational Inference*

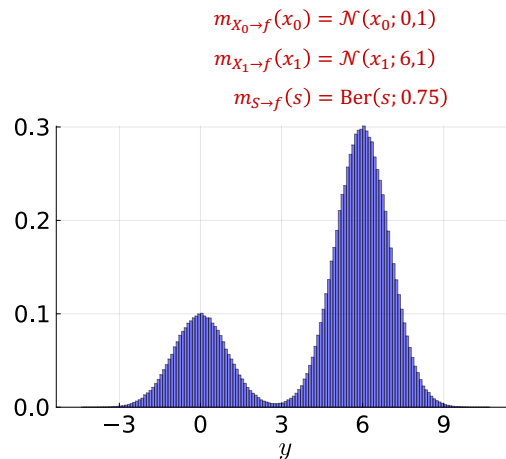
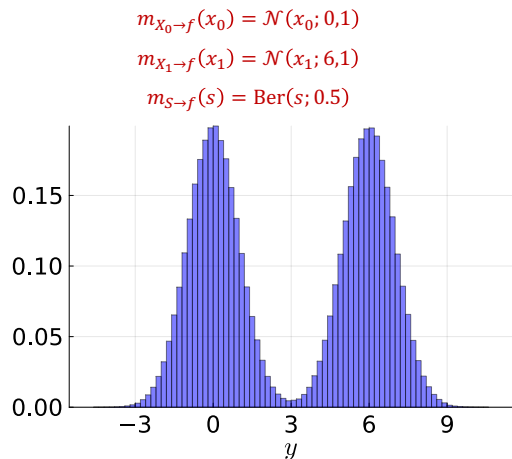
Motivation: Mixture Models

- **Mixture Model:** If we have two possible causes X_0 and X_1 for an observed effect Y and a (non-observed) binary selector $S \in \{0,1\}$ then

$$f(X_0, X_1, S, Y) = \sum_{s=0}^1 \mathbb{I}(S = s) \cdot \delta(Y - X_s) = \delta(Y - X_S)$$



- **Example:** A match outcome model where the skill of a player is a mixture of the skills of two real people in a household



```
# implements the mixture factor
function mixture(x0, x1, s)
    return (s == 0) ? x0 : x1
end

# samples the message to the output variable for a mixture factor
function sample_mixture_message()
    mu_x0 = 0.0, sigma_x0=1.0, mu_x1=6.0, sigma_x1=1.0, p=0.5,
    n_samples=1000000
)

# stores the samples of y
samples_msg_to_y = Vector{Float64}(undef, n_samples)
for i in 1:n_samples
    x0 = randn() * sigma_x0 + mu_x0
    x1 = randn() * sigma_x1 + mu_x1
    s = rand() < p ? 1 : 0
    # compute y using the mixture factor
    samples_msg_to_y[i] = mixture(x0, x1, s)
end
end
```

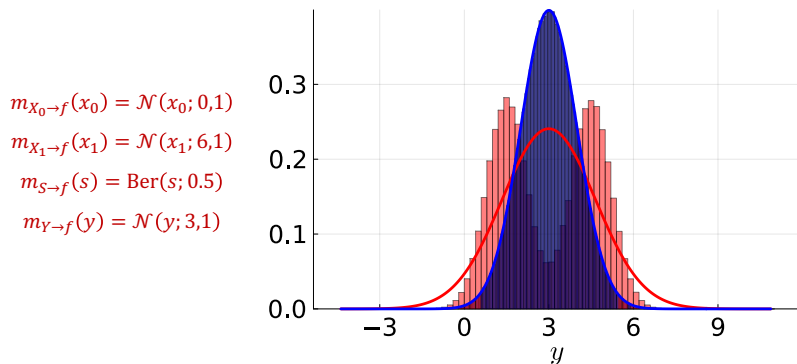
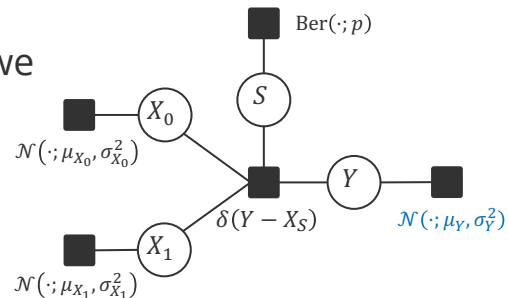
Motivation: Mixture Models (ctd)

- **Expectation Propagation:** Since the message $m_{f \rightarrow Y}(\cdot)$ is not a Gaussian, we approximate the marginal $p_Y(\cdot) = m_{f \rightarrow Y}(\cdot) \cdot m_{Y \rightarrow f}(\cdot)$ with $\hat{p}_Y(\cdot)$ by virtue of

$$\hat{p}_Y(\cdot) = \mathcal{N}(\cdot; \underbrace{E_{Y \sim p_Y}[Y]}_{\hat{\mu}_Y}, \underbrace{\text{var}_{Y \sim p_Y}[Y]}_{\hat{\sigma}_Y^2})$$

- **Idea:** Instead of algebraic integration, use importance sampling with the proposal

$$q(x_0, x_1, s, y) = \mathcal{N}(x_0; \mu_0, \sigma_0^2) \cdot \mathcal{N}(x_1; \mu_1, \sigma_1^2) \cdot \text{Ber}(s; p) \cdot \delta(y - x_s) \quad \Rightarrow \quad w(x_0, x_1, s, y) = \frac{\prod_f f(x_0, x_1, s, y)}{q(x_0, x_1, s, y)} = \mathcal{N}(y; \mu_Y, \sigma_Y^2)$$



$$\hat{m}_{f \rightarrow Y}(y) = \frac{\hat{p}_Y(y)}{m_{Y \rightarrow f}(y)}$$

$$= \frac{\mathcal{N}(y; \hat{\mu}_Y, \hat{\sigma}_Y^2)}{\mathcal{N}(y; \mu_Y, \sigma_Y^2)}$$

Not possible because $\hat{\sigma}_Y^2 > \sigma_Y^2$

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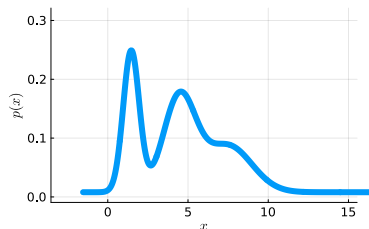
Recall: α -Divergence

- **α -Divergence (Amari, 1985).** Given two probability densities $p(\cdot)$ and $q(\cdot)$ and $\alpha \in \mathbb{R} \setminus \{0,1\}$ the α -divergence $D_\alpha[p, q]$ is defined by

$$D_\alpha[p, q] = \frac{1}{\alpha(1-\alpha)} \cdot \left(1 - \int_{-\infty}^{+\infty} \left[\frac{p(x)}{q(x)} \right]^\alpha \cdot q(x) dx \right)$$



Shun'ichi Amari (甘利 俊)
(1936)



zero-forcing & mode seeking

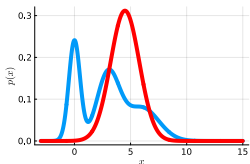
inclusive & support seeking

α

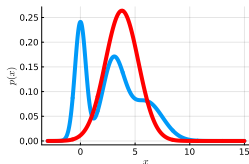
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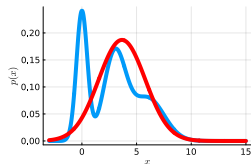
$\alpha = -\infty$



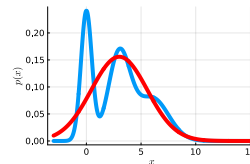
$\alpha = -1$



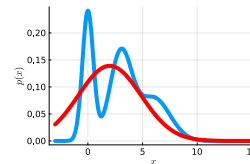
$\alpha = 0$



$\alpha = 1$



$\alpha = +\infty$



$\alpha = 0$: Reverse KL Divergence

- **Theorem (Limit $\alpha \rightarrow 0$).** Given two probability densities $p(\cdot)$ and $q(\cdot)$ the limit of the α -divergence $D_\alpha[p, q]$ for $\alpha \rightarrow 0$ is the reverse Kullback-Leibler divergence

Note that for $\alpha \rightarrow 1$ we get $\text{KL}[p(\cdot), q(\cdot)]$

$$\lim_{\alpha \rightarrow 0} D_\alpha[p, q] \Rightarrow \text{KL}[q(\cdot), p(\cdot)] := \int_{-\infty}^{+\infty} \log\left(\frac{q(x)}{p(x)}\right) \cdot q(x) \, dx$$

- **Proof:** Taking limits, we have

$$\begin{aligned} \lim_{\alpha \rightarrow 0} D_\alpha[p, q] &= \lim_{\alpha \rightarrow 0} \frac{1}{\alpha(1-\alpha)} \cdot \left(1 - \int_{-\infty}^{+\infty} \left[\frac{p(x)}{q(x)} \right]^\alpha \cdot q(x) \, dx \right) \quad \text{L'Hôpital's rule!} \\ &= \lim_{\alpha \rightarrow 0} \frac{1}{1-2\alpha} \cdot \left(- \int_{-\infty}^{+\infty} \log\left(\frac{p(x)}{q(x)}\right) \cdot \left[\frac{p(x)}{q(x)} \right]^\alpha \cdot q(x) \, dx \right) \quad \text{Note that } \frac{d}{dz} b^z = \log(b) \cdot b^z \\ &= \int_{-\infty}^{+\infty} \log\left(\frac{q(x)}{p(x)}\right) \cdot q(x) \, dx \end{aligned}$$

- **Challenge:** Given an arbitrary density $p(\cdot)$ how do we efficiently find the minimizer of the reverse Kullback-Leibler divergence when

$$q(\cdot) = \exp\left(\boldsymbol{\theta}^T \mathbf{T}(\cdot) - A(\boldsymbol{\theta})\right) \quad \text{Exponential family parameterized by } \mathbf{T}(\cdot)$$



Solomon Kullback
(1909 – 1994)



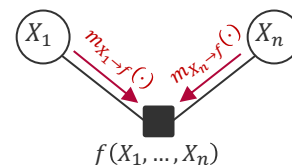
Richard Leibler
(1914 – 2003)

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Evidence Lower Bound (ELBO)

- Consider a single factor f of n non-observed variables $\mathbf{X} := X_1, \dots, X_n$
 - All other non-observed variables $X_{n+1}, X_{n+2}, \dots$ have already been integrated out
 - All observed variables (i.e., data D) are encoded through other factors and f



$$p(\mathbf{x}, D) = f(\mathbf{x}_1, \dots, \mathbf{x}_n) \cdot \prod_{i=1}^n m_{X_i \rightarrow f}(x_i)$$

- **Observation 1:** Since the data D is observed, we know that $p(D)$ is constant
- **Observation 2:** For any distribution q over $X_1, \dots, X_n$ we know that

$$\begin{aligned} \log(p(D)) &= \log(p(D)) \cdot \int q(\mathbf{x}) \, d\mathbf{x} \quad \leftarrow = 1 \\ &= \int q(\mathbf{x}) \cdot \left[\log\left(\frac{p(\mathbf{x}, D)}{p(\mathbf{x}|D)} \cdot \frac{q(\mathbf{x})}{q(\mathbf{x})}\right) \right] d\mathbf{x} \quad \leftarrow p(\mathbf{x}|D) = \frac{p(\mathbf{x}, D)}{p(D)} \quad \leftarrow = 1 \\ &= \int q(\mathbf{x}) \cdot \left[\log\left(\frac{p(\mathbf{x}, D)}{q(\mathbf{x})}\right) \right] d\mathbf{x} + \int q(\mathbf{x}) \cdot \left[\log\left(\frac{q(\mathbf{x})}{p(\mathbf{x}|D)}\right) \right] d\mathbf{x} \end{aligned}$$

ELBO[$q(\cdot), p(\cdot, D)$]

KL[$q(\cdot), p(\cdot | D)$] ≥ 0

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Evidence Lower Bound (ELBO): Example

$$\text{KL}[q(\cdot), p(\cdot | D)] = \log(p(D)) + \text{ELBO}[q(\cdot), p(\cdot, D)] \geq 0$$

$$\log(p(D)) \geq \text{ELBO}[q(\cdot), p(\cdot, D)]$$

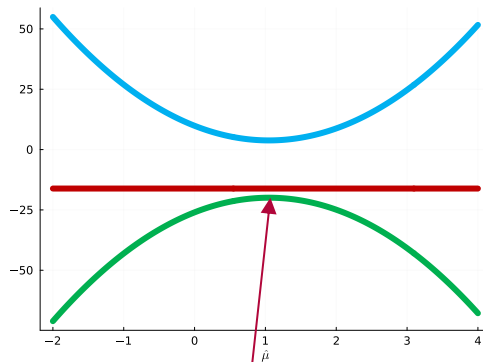
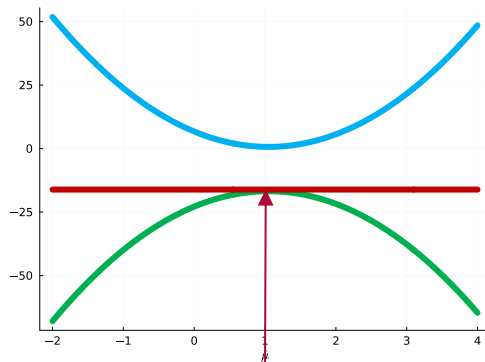
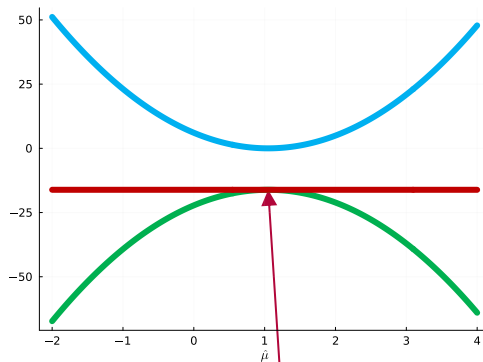
$\text{ELBO}[q(\cdot), p(\cdot, D)]$ is a lower bound of the (constant) evidence $\log(p(D))$

$$p(X, D) = \mathcal{N}(X; 0, 1) \cdot \prod_{i=1}^{10} \mathcal{N}(x_i; X, 1)$$

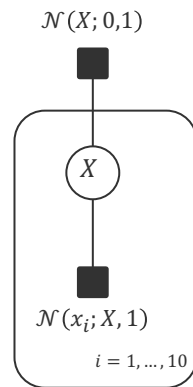
$$q(\cdot) = \mathcal{N}\left(\cdot; \hat{\mu}, \frac{1}{11}\right)$$

$$q(\cdot) = \mathcal{N}\left(\cdot; \hat{\mu}, \frac{1}{100}\right)$$

$$q(\cdot) = \mathcal{N}(\cdot; \hat{\mu}, 1)$$



Maximizing $\text{ELBO}[q(\cdot), p(\cdot, D)]$ over $q(\cdot)$ finds the minimizer of $\text{KL}[q(\cdot), p(\cdot | D)]$



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ELBO for Factorizing Distributions

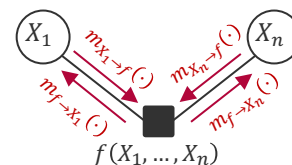
- **Assumption:** Since we are only interested in the marginals $p_{X_i}(\cdot)$ we will assume

$$q(x_1, \dots, x_n) = \prod_{i=1}^n q_i(x_i)$$

- For our factor graph model, we have $q_i(x_i) \propto m_{f \rightarrow X_i}(x_i) \cdot m_{X_i \rightarrow f}(x_i)$

- In this case the ELBO function becomes a sum of two terms

$$\begin{aligned} \text{ELBO}[q(\cdot), p(\cdot, D)] &= \int q(\mathbf{x}) \cdot \left[\log \left(\frac{p(\mathbf{x}, D)}{q(\mathbf{x})} \right) \right] d\mathbf{x} \\ &= \int \left[\prod_{i=1}^n q_i(x_i) \right] \cdot \left[\log \left(\frac{p(\mathbf{x}, D)}{\prod_{i=1}^n q_i(x_i)} \right) \right] d\mathbf{x} \\ &= \int \left[\prod_{i=1}^n q_i(x_i) \right] \cdot \log(p(\mathbf{x}, D)) d\mathbf{x} - \sum_{i=1}^n \int q_i(x_i) \cdot \log(q_i(x_i)) dx_i \end{aligned}$$



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Maximizing ELBO for Factorizing Distributions

- Idea:** Assuming only $q_j(\cdot)$ is unknown, we derive the maximizer of ELBO.

$$\begin{aligned}
 \text{ELBO}[q_j(\cdot), p(\cdot, D)] &= \int \left[\prod_{i=1}^n q_i(x_i) \right] \cdot \log(p(\mathbf{x}, D)) \, d\mathbf{x} - \sum_{i=1}^n \int q_i(x_i) \cdot \log(q_i(x_i)) \, dx_i \\
 &= \int q_j(x_j) \cdot \left(\int \left[\prod_{i \neq j}^n q_i(x_i) \right] \cdot \log(p(\mathbf{x}, D)) \, d\mathbf{x}_{\setminus j} \right) dx_j - \int q_j(x_j) \cdot \log(q_j(x_j)) \, dx_j + \text{const} \\
 &= - \int q_j(x_j) \cdot \log \left(\frac{q_j(x_j)}{\exp(E_j(x_j))} \right) dx_j + \text{const}
 \end{aligned}$$

$q(x_1, \dots, x_n) = q_j(x_j) \cdot \prod_{i \neq j}^n q_i(x_i)$
 (The product term is labeled "const" with an arrow pointing to it.)

$E_j(x_j)$ where all X_i with $i \neq j$ are integrated out

$\text{KL}[q(\cdot), \exp(E_j(\cdot))] \geq 0$ with equality if and only if $q(\cdot) \propto \exp(E_j(\cdot))$

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- Thus, the ELBO maximiser $q_j^*(\cdot)$ must fulfill the stationary condition:

$$q_j^*(x_j) \propto \exp(E_j(x_j))$$

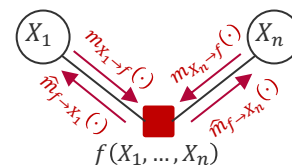
Note that the $E_j(\cdot)$ requires all other $q_i(\cdot)$ so these are n coupled conditions for the ELBO maximiser!

Variational Inference for Exponential Family

- Consider a single factor f of n non-observed variables $\mathbf{X} := X_1, \dots, X_n$

$$p(\mathbf{x}, D) = f(x_1, \dots, x_n) \cdot \prod_{i=1}^n m_{X_i \rightarrow f}(x_i)$$

- Assumption 1:** All incoming messages from the variables X_i are in the exponential family parameterized by the sufficient statistic $\mathbf{T}_i(\cdot)$



$$m_{X_i \rightarrow f}(x_i) \propto \exp(\boldsymbol{\theta}_i^T \mathbf{T}_i(x_i))$$

- Assumption 2:** The factor f has the n properties that for all $i = 1, \dots, n$

$$f(x_1, \dots, x_n) \propto \exp(\mathbf{O}_i(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)^T \mathbf{T}_i(x_i))$$

$$q_i^*(x_i) \propto \exp\left(\int \left[\prod_{j \neq i}^n q_j(x_j)\right] \cdot \log(p(\mathbf{x}, D)) \, d\mathbf{x}_{\setminus i}\right)$$

- Then, the variational approximation $q_i^*(\cdot)$ of the marginal $p_{X_i}(\cdot)$ is

$$q_i^*(x_i) \propto \exp\left(\boldsymbol{\theta}_i^T \mathbf{T}_i(x_i) + E_{X_1 \sim q_1^*, \dots, X_{i-1} \sim q_{i-1}^*, X_{i+1} \sim q_{i+1}^*, \dots, X_n \sim q_n^*}[\mathbf{O}_i(X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n)]^T \mathbf{T}_i(x_i)\right)$$

- Dividing out $m_{X_i \rightarrow f}(x_i)$, the variational approximation $\hat{m}_{f \rightarrow X_i}(\cdot)$ is given by

$$\hat{m}_{f \rightarrow X_i}(x_i) \propto \exp\left(E_{X_1 \sim q_1^*, \dots, X_{i-1} \sim q_{i-1}^*, X_{i+1} \sim q_{i+1}^*, \dots, X_n \sim q_n^*}[\mathbf{O}_i(X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n)]^T \mathbf{T}_i(x_i)\right)$$

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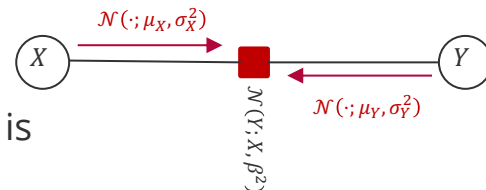
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Gaussian Mean Factor: Joint Distribution

$$f(X, Y) = \mathcal{N}(Y; X, \beta^2)$$



- The true **joint distribution** given this factor and the incoming messages is

$$p(x, y) = \mathcal{N}(x; \mu_X, \sigma_X^2) \cdot \mathcal{N}(y; \mu_Y, \sigma_Y^2) \cdot \mathcal{N}(y; x, \beta^2)$$

$$\propto \mathcal{N} \left(\begin{bmatrix} x \\ y \end{bmatrix}; \begin{bmatrix} \mu_X \cdot \frac{\sigma_Y^2 + \beta^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2} + \mu_Y \cdot \frac{\sigma_X^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2} \\ \mu_X \cdot \frac{\sigma_Y^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2} + \mu_Y \cdot \frac{\sigma_X^2 + \beta^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2} \end{bmatrix}, \begin{bmatrix} \sigma_X^2 \cdot \frac{\sigma_Y^2 + \beta^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2} & \sigma_X^2 \sigma_Y^2 \cdot \frac{1}{\sigma_X^2 + \sigma_Y^2 + \beta^2} \\ \sigma_X^2 \sigma_Y^2 \cdot \frac{1}{\sigma_X^2 + \sigma_Y^2 + \beta^2} & \sigma_Y^2 \cdot \frac{\sigma_X^2 + \beta^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2} \end{bmatrix} \right)$$

Weighted average of the means with weights proportional to the variance ratios

Scaled (co)variances with weights proportional to the variance ratios

- Derived using the natural parameterization and collecting terms for x , y , x^2 , y^2 and $x \cdot y$
- In the limit case of $\beta \rightarrow \infty$, the factor $\mathcal{N}(y; x, \beta^2)$ becomes the uniform-Normal and

$$p(x, y) \propto \mathcal{N} \left(\begin{bmatrix} x \\ y \end{bmatrix}; \begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix}, \begin{bmatrix} \sigma_X^2 & 0 \\ 0 & \sigma_Y^2 \end{bmatrix} \right)$$

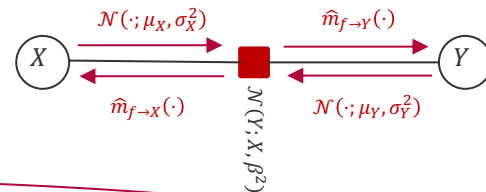
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Gaussian Mean Factor: Variational Approximation

$$\begin{aligned}
 f(X, Y) &= \mathcal{N}(Y; X, \beta^2) \\
 &= \mathcal{G}(Y; \beta^{-2}X, \beta^{-2}) \propto \exp \left(\begin{bmatrix} \beta^{-2}X \\ \beta^{-2} \end{bmatrix}^T \begin{bmatrix} Y \\ -\frac{Y^2}{2} \end{bmatrix} \right) \\
 &= \mathcal{G}(X; \beta^{-2}Y, \beta^{-2}) \propto \exp \left(\begin{bmatrix} \beta^{-2}Y \\ \beta^{-2} \end{bmatrix}^T \begin{bmatrix} X \\ -\frac{X^2}{2} \end{bmatrix} \right)
 \end{aligned}$$

Annotations: $\mathbf{o}_Y(X)$ points to $\beta^{-2}X$, $\mathbf{T}_Y(Y)$ points to $\begin{bmatrix} Y \\ -\frac{Y^2}{2} \end{bmatrix}$, $\mathbf{o}_X(Y)$ points to $\beta^{-2}Y$, $\mathbf{T}_X(X)$ points to $\begin{bmatrix} X \\ -\frac{X^2}{2} \end{bmatrix}$.



$$\begin{aligned}
 \hat{m}_{f \rightarrow X}(x) &\propto \exp \left(E_{Y \sim \hat{p}_Y} [\mathbf{o}_X(Y)]^T \mathbf{T}_X(x) \right) \propto \mathcal{G}(x; \beta^{-2} E_{Y \sim \hat{p}_Y} [Y], \beta^{-2}) \\
 \hat{m}_{f \rightarrow Y}(y) &\propto \exp \left(E_{X \sim \hat{p}_X} [\mathbf{o}_Y(X)]^T \mathbf{T}_Y(y) \right) \propto \mathcal{G}(y; \beta^{-2} E_{X \sim \hat{p}_X} [X], \beta^{-2})
 \end{aligned}$$

$$\hat{p}_Y(y) = \mathcal{N}(y; \mu_Y, \sigma_Y^2) \cdot \hat{m}_{f \rightarrow Y}(y)$$

$$\hat{p}_X(x) = \mathcal{N}(x; \mu_X, \sigma_X^2) \cdot \hat{m}_{f \rightarrow X}(x)$$

- The two messages are coupled due to the expectation of X and Y !
- We can analytically solve these two equations for the joint solution to obtain

$$\hat{p}(x, y) \propto \mathcal{N} \left(\begin{bmatrix} x \\ y \end{bmatrix}; \begin{bmatrix} \mu_X \cdot \frac{\sigma_Y^2 + \beta^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2} + \mu_Y \cdot \frac{\sigma_X^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2} \\ \mu_X \cdot \frac{\sigma_Y^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2} + \mu_Y \cdot \frac{\sigma_X^2 + \beta^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2} \end{bmatrix}, \begin{bmatrix} \sigma_X^2 \cdot \frac{\beta^2}{\sigma_X^2 + \beta^2} & 0 \\ 0 & \sigma_Y^2 \cdot \frac{\beta^2}{\sigma_Y^2 + \beta^2} \end{bmatrix} \right)$$

Annotations: Red boxes highlight $\sigma_X^2 \cdot \frac{\beta^2}{\sigma_X^2 + \beta^2}$ and $\sigma_Y^2 \cdot \frac{\beta^2}{\sigma_Y^2 + \beta^2}$. A red arrow points from the text 'Smaller than the true variance...' to these terms.

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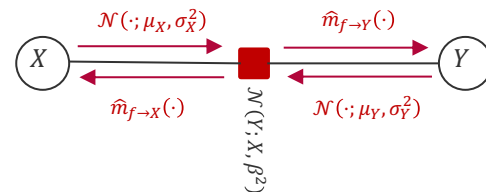
Smaller than the true
variance of $\sigma_X^2 \cdot \frac{\sigma_Y^2 + \beta^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2}$

Gaussian Mean Factor: Examples

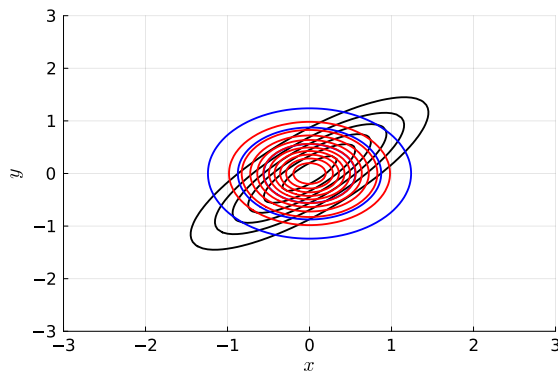
$$m_{X \rightarrow f}(x) = \mathcal{N}(x; 0, 1)$$

$$m_{Y \rightarrow f}(y) = \mathcal{N}(y; 0, 1)$$

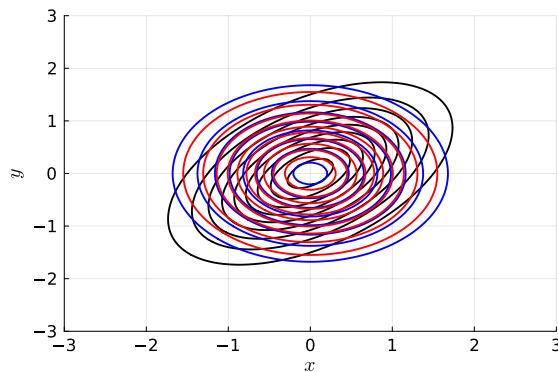
$$f(x, y) = \mathcal{N}(y; x, \beta^2)$$



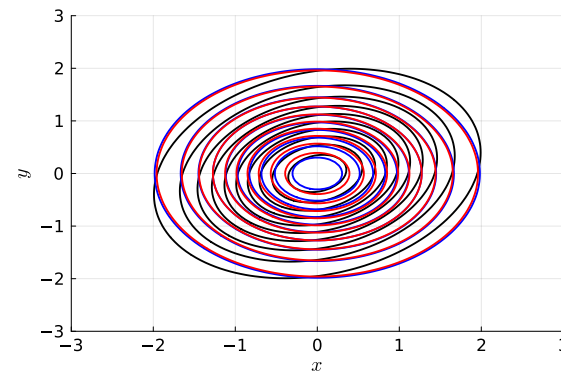
$\beta = 0.5$



$\beta = 1$



$\beta = 2$



Gaussian Factor with Variable Precision

$$f(X, S) = \mathcal{N}(X; y, S^{-1})$$

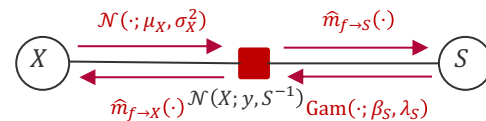
$$= \exp\left(\frac{1}{2}\log(S) - \frac{1}{2}\log(2\pi) - \frac{S \cdot (X - y)^2}{2}\right)$$

$$\propto \exp\left(\begin{bmatrix} S \cdot y \\ S \end{bmatrix}^T \begin{bmatrix} X \\ X^2 \\ -\frac{1}{2} \end{bmatrix}\right)$$

$$\propto \exp\left(\begin{bmatrix} \frac{1}{2} \\ \frac{(X - y)^2}{2} \end{bmatrix}^T \begin{bmatrix} \log(S) \\ -S \end{bmatrix}\right)$$

$$\hat{m}_{f \rightarrow X}(x) \propto \exp\left(E_{S \sim \hat{p}_S}[\mathbf{O}_X(S)]^T \mathbf{T}_X(x)\right) \\ \propto \mathcal{G}(x; y \cdot E_{S \sim \hat{p}_S}[S], E_{S \sim \hat{p}_S}[S])$$

$$\hat{m}_{f \rightarrow S}(s) \propto \exp\left(E_{X \sim \hat{p}_X}[\mathbf{O}_S(X)]^T \mathbf{T}_S(s)\right) \\ \propto \text{Gam}\left(s; \frac{1}{2}, \frac{E_{X \sim \hat{p}_X}[X^2] + y^2}{2} - y \cdot E_{X \sim \hat{p}_X}[X]\right)$$



$$\hat{p}_S(s) = \text{Gam}(s; \beta_S, \lambda_S) \cdot \hat{m}_{f \rightarrow S}(s)$$

$$\hat{p}_X(x) = \mathcal{N}(x; \mu_X, \sigma_X^2) \cdot \hat{m}_{f \rightarrow X}(x)$$

**Advanced Probabilistic
Machine Learning**

Unit 6 – Approximate Inference:
Variational Inference

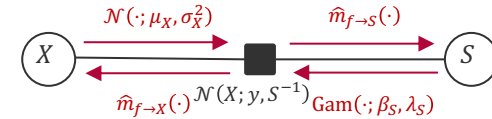
- The two messages are coupled due to the expectation of X and S !
- There is no analytical solution to solve them simultaneously
- Thus, they get iterated until convergence from a start-point given by the incoming messages $\mathcal{N}(\cdot; \mu_X, \sigma_X^2)$ and $\text{Gam}(\cdot; \beta_S, \lambda_S)$

Gaussian Factor with Variable Precision: Examples

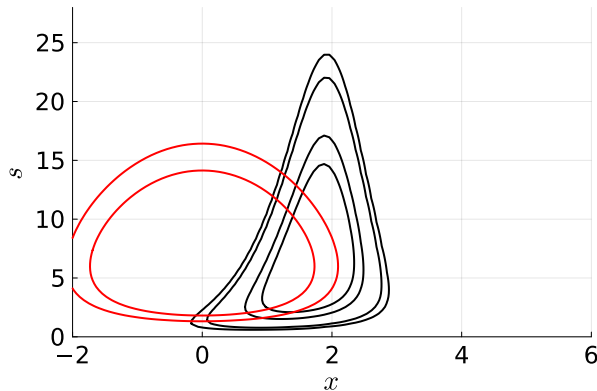
$$m_{X \rightarrow f}(x) = \mathcal{N}(x; 0, 1)$$

$$m_{S \rightarrow f}(s) = \text{Gam}(s; 3, 0.5)$$

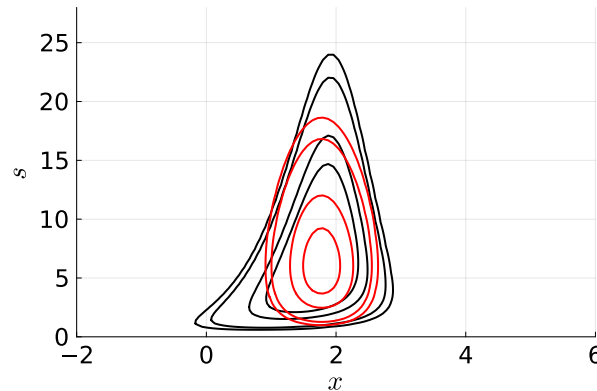
$$f(x, s) = \mathcal{N}(x; y, s^{-1})$$



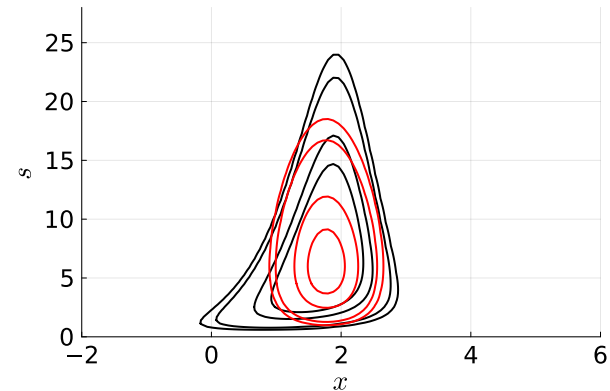
Start



Iteration 1



Iteration 2



Gaussian Mean Factor with Variable Precision

$$f(X, Y, S) = \mathcal{N}(X; Y, S^{-1}) = \exp\left(\frac{1}{2}\log(S) - \frac{1}{2}\log(2\pi) - \frac{S \cdot (X - Y)^2}{2}\right)$$

$$\mathbf{o}_X(S, Y) \propto \exp\left(\begin{bmatrix} S \cdot Y \\ S \end{bmatrix}^T \begin{bmatrix} X \\ -\frac{X^2}{2} \end{bmatrix}\right) \mathbf{T}_X(X)$$

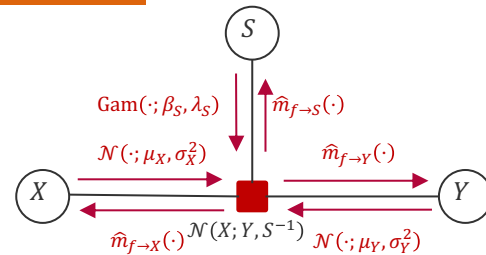
$$\mathbf{o}_Y(S, X) \propto \exp\left(\begin{bmatrix} S \cdot X \\ S \end{bmatrix}^T \begin{bmatrix} Y \\ -\frac{Y^2}{2} \end{bmatrix}\right) \mathbf{T}_Y(Y)$$

$$\mathbf{o}_S(X, Y) \propto \exp\left(\begin{bmatrix} \frac{1}{2} \\ \frac{(X - Y)^2}{2} \\ -S \end{bmatrix}^T \begin{bmatrix} \log(S) \\ -S \end{bmatrix}\right) \mathbf{T}_S(S)$$

$$\begin{aligned} \hat{m}_{f \rightarrow X}(x) &\propto \exp\left(E_{Y \sim \hat{p}_Y, S \sim \hat{p}_S}[\mathbf{o}_X(S, Y)]^T \mathbf{T}_X(x)\right) \\ &\propto \mathcal{G}(x; E_{Y \sim \hat{p}_Y}[Y] \cdot E_{S \sim \hat{p}_S}[S], E_{S \sim \hat{p}_S}[S]) \end{aligned}$$

$$\begin{aligned} \hat{m}_{f \rightarrow Y}(y) &\propto \exp\left(E_{X \sim \hat{p}_X, S \sim \hat{p}_S}[\mathbf{o}_Y(S, X)]^T \mathbf{T}_Y(y)\right) \\ &\propto \mathcal{G}(y; E_{X \sim \hat{p}_X}[X] \cdot E_{S \sim \hat{p}_S}[S], E_{S \sim \hat{p}_S}[S]) \end{aligned}$$

$$\begin{aligned} \hat{m}_{f \rightarrow S}(s) &\propto \exp\left(E_{X \sim \hat{p}_X, Y \sim \hat{p}_Y}[\mathbf{o}_S(X, Y)]^T \mathbf{T}_S(s)\right) \\ &\propto \text{Gam}\left(s; \frac{1}{2}, \frac{E_{X \sim \hat{p}_X}[X^2] + E_{Y \sim \hat{p}_Y}[Y^2]}{2} - E_{Y \sim \hat{p}_Y}[Y] \cdot E_{X \sim \hat{p}_X}[X]\right) \end{aligned}$$



$$\hat{p}_S(s) = \text{Gam}(s; \beta_S, \lambda_S) \cdot \hat{m}_{f \rightarrow S}(s)$$

$$\hat{p}_Y(y) = \mathcal{N}(y; \mu_Y, \sigma_Y^2) \cdot \hat{m}_{f \rightarrow Y}(y)$$

$$\hat{p}_X(x) = \mathcal{N}(x; \mu_X, \sigma_X^2) \cdot \hat{m}_{f \rightarrow X}(x)$$

Advanced Probabilistic
Machine Learning

Unit 6 – Approximate Inference:
Variational Inference

- The three messages (six equations) are coupled due to the expectation of X , Y and S
- Iterate to solve for the six parameters of the messages $\hat{m}_{f \rightarrow X}(\cdot)$, $\hat{m}_{f \rightarrow Y}(\cdot)$ and $\hat{m}_{f \rightarrow S}(\cdot)$

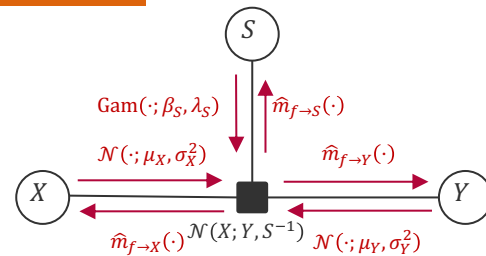
Gaussian Factor with Variable Precision: Examples

$$m_{X \rightarrow f}(x) = \mathcal{N}(x; 0, 1)$$

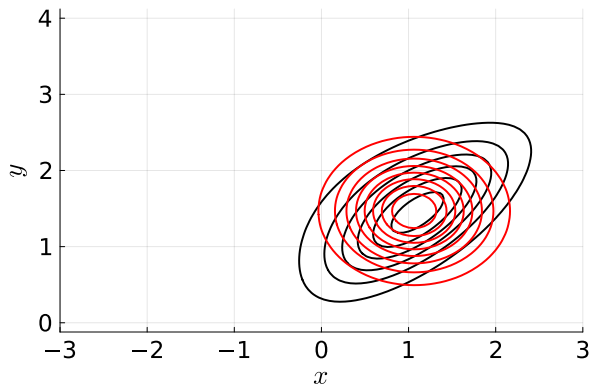
$$m_{Y \rightarrow f}(y) = \mathcal{N}(y; 2, 0.5)$$

$$m_{S \rightarrow f}(s) = \text{Gam}(s; 2, 1)$$

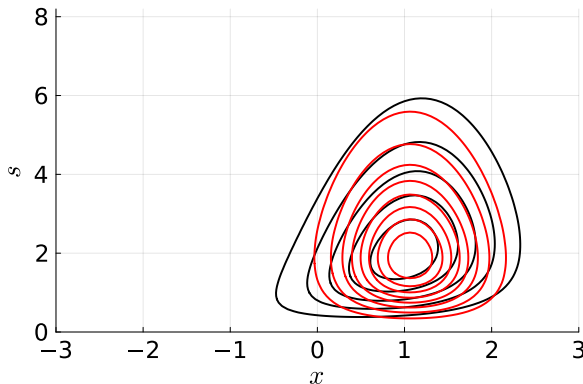
$$f(x, y, s) = \mathcal{N}(x; y, s^{-1})$$



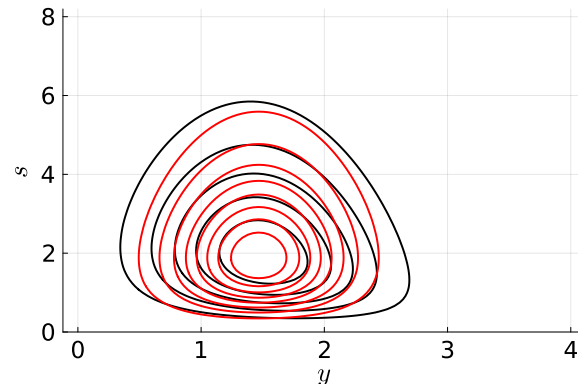
Marginal p_{XY}



Marginal p_{XS}



Marginal p_{YS}



Overview

1. Minimizing Reverse KL-Divergence via ELBO
2. Variational Inference for Gaussian Factors
3. **Application: Robust Bayesian Regression and Ranking**

**Advanced Probabilistic
Machine Learning**

*Unit 6 – Approximate Inference:
Variational Inference*

Robust Bayesian Regression

■ Given:

1. **Training Data:** $D \in (\mathcal{X} \times \mathbb{R})^n$ of n (labelled) examples (x_i, y_i)
2. **Linear Basis Functions:** Basis function mapping $\phi: \mathcal{X} \rightarrow \mathbb{R}^M$ and linear function model $f(x; \mathbf{w}):= \mathbf{w}^T \phi(x)$
3. **Per-Data Point Precisions** $s_i \in \mathbb{R}^+$
4. **Likelihood of functions and precisions:**

$$p(D|\mathbf{w}, \mathbf{s}) = \prod_{i=1}^n \mathcal{N}(y_i; \mathbf{w}^T \phi(x_i), s_i^{-1})$$

5. **Prior belief over functions and precisions:**

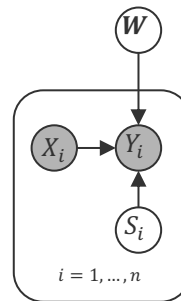
$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$$

$$p(\mathbf{s}) = \prod_{i=1}^n \text{Gam}(s_i; \beta, \lambda)$$

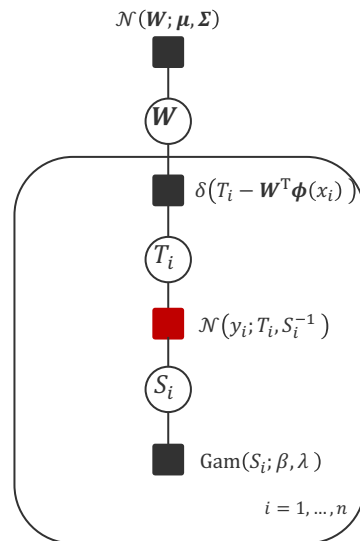
- ## ■ Bayesian Inference:
- Posterior belief over functions and precisions

$$p(\mathbf{w}, \mathbf{s}|D) \propto \left[\prod_{i=1}^n \mathcal{N}(y_i; \mathbf{w}^T \phi(x_i), s_i^{-1}) \cdot \text{Gam}(s_i; \beta, \lambda) \right] \cdot \mathcal{N}(\mathbf{w}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$$

Bayesian Network



Factor Graph

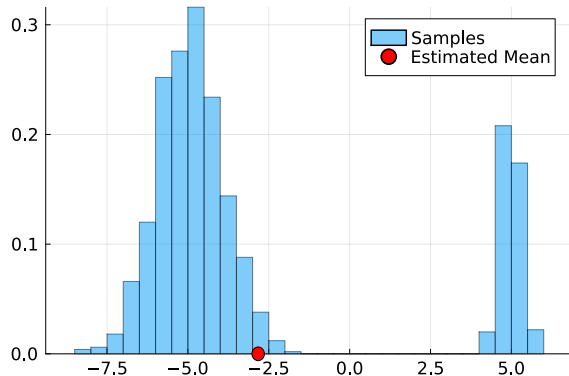


Unit 6 – Approximate Inference:
Variational Inference

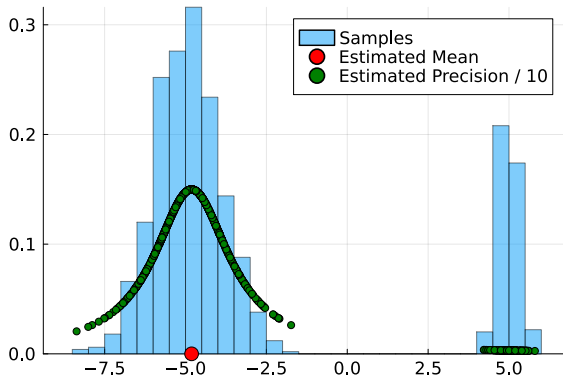
Robust Bayesian Regression: Warm-Up

- **Simple Model:** We have no features so W is a single number T (the average of the observations y_i)
- **Message Schedule:**
 1. Send exact prior messages to T and all S_i
 2. For each observation y_i , update variational message to S_i
 3. For each observation y_i , update variational message to T

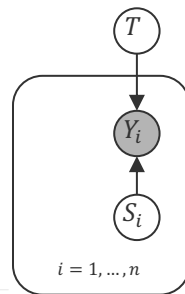
Fixed noise ($\beta^2 = 0.01$)



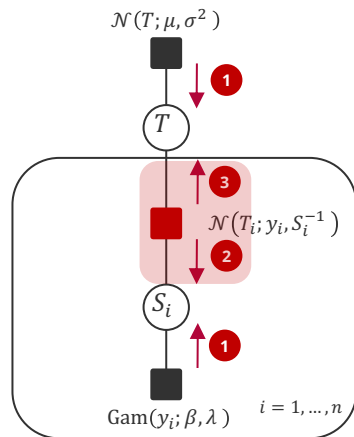
Variable noise



Bayesian Network



Factor Graph



Advanced Probabilistic Machine Learning

Unit 6 – Approximate Inference:
Variational Inference

$n = 1000$

$\mu = 0$

$\sigma^2 = 1$

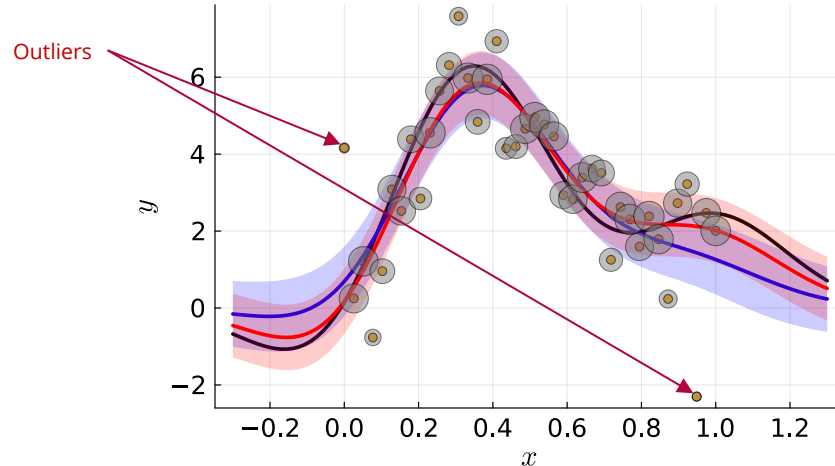
$\beta = 0$

$\lambda = 1$

Robust Bayesian Regression

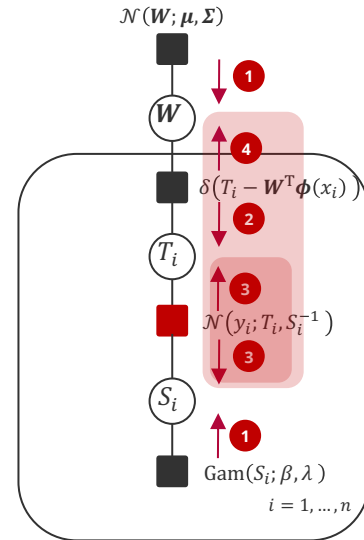
■ Message Schedule:

1. Send exact prior messages to $\mathbf{W}$ and all S_i
- Outer loop iterating over the observations (x_i, y_i) :
 2. Send the message from the projection factor to T_i
 3. Iterate the variational messages from the Gaussian factor to S_i and T_i
 4. Send the message from the projection factor to $\mathbf{W}$



$$\begin{aligned}\mu &= 0 \\ \Sigma &= I \\ \phi(x) &= \begin{bmatrix} \mathcal{N}(x; 0, 0.2^2) \\ \vdots \\ \mathcal{N}(x; 1, 0.2^2) \end{bmatrix} \\ \beta &= 0.5 \\ \lambda &= 1 \\ n &= 40\end{aligned}$$

Factor Graph



Unit 6 – Approximate Inference:
Variational Inference

Robust Bayesian Ranking

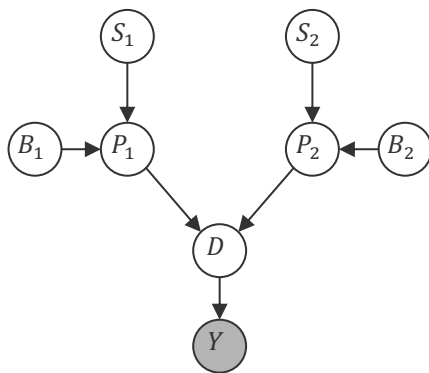
■ Classic:

- Two players have unknown skills S_1 and S_2 with Gaussian belief
- Performance P_i in a match is Gaussian with mean S_i
- Match-outcome $y \in \{-1, +1\}$ means that $y \cdot (P_1 - P_2) > 0$

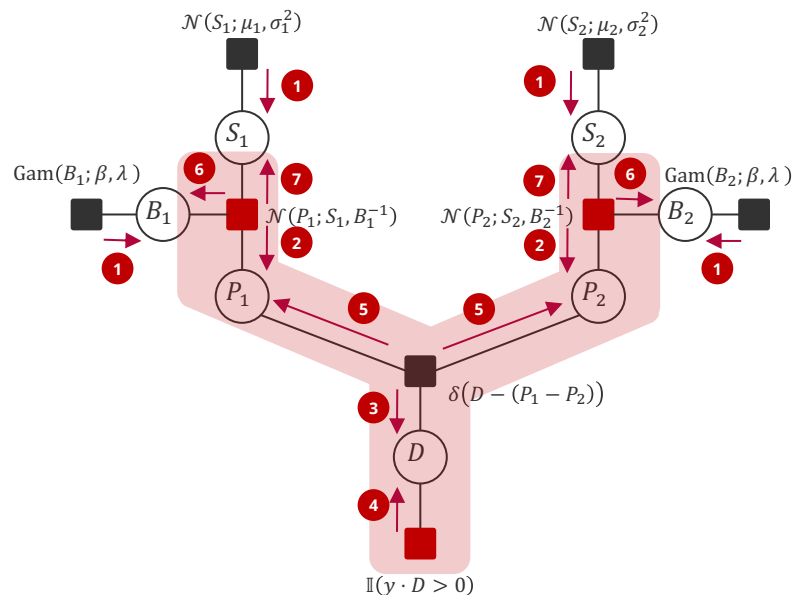
■ Robust:

- We assume an unknown performance variation B_i with Gamma distribution
- Performance P_i in a match is Gaussian with mean S_i and variance B_i^{-1}
- The rest is the same!

Bayesian Network



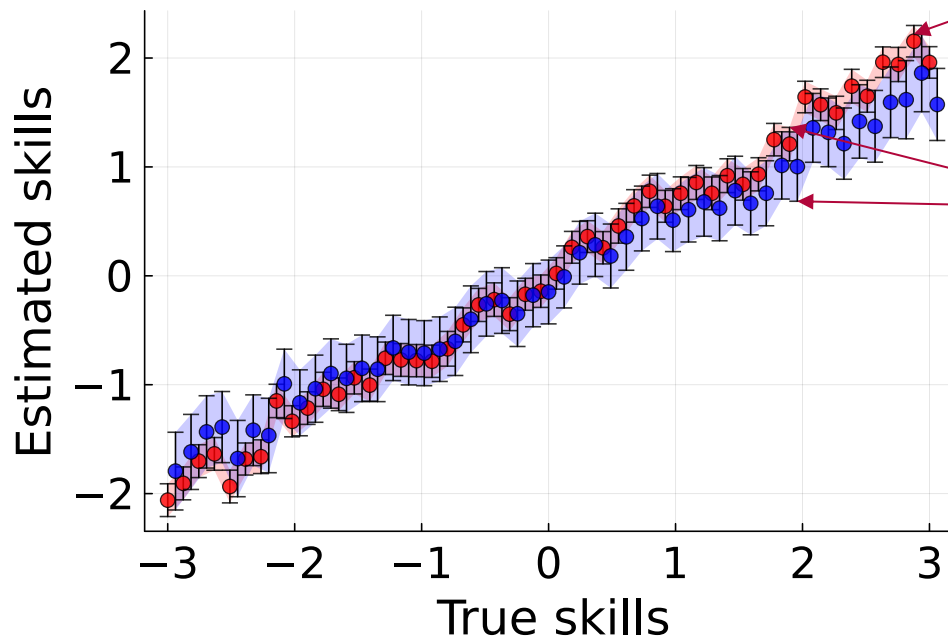
Factor Graph



Robust Bayesian Ranking: Example

■ Experimental Setup:

- We have 50 players with skills evenly distributed in $[-3,3]$
- The prior of the performance precision is $B_i \sim \text{Gam}(0,1)$
- We generate 10,000 matches with random player indices and outcomes using random performance precisions
- We ran TrueSkill Through Time both with fixed (blue) and variable (red) performance precision



Variable performance precision model can disregard extreme match outcomes and models the full skill range

Variable performance precision model gets more certain skill estimates from the same data

Advanced Probabilistic Machine Learning

Unit 6 – Approximate Inference: Variational Inference

Robust Ranking: Modelling Performance Precisions

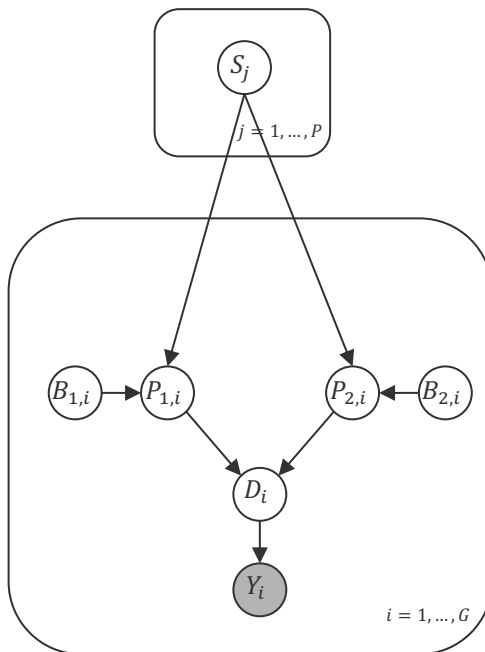
■ Per-Match Precisions:

- Number of skills variables equals number P of players
- Number of performance precisions equals number G of games

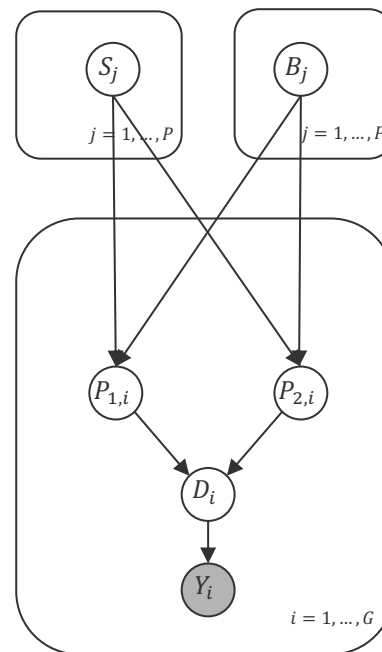
■ Per-Player Precisions:

- Number of skills variables equals number P of players
- Number of performance precisions equals number P of players!
- **Pro:** Per-player precisions model a property of players
- **Con:** Loopy graph leads to over-confidence

Per-Match Performance Precisions



Per-Player Performance Precisions

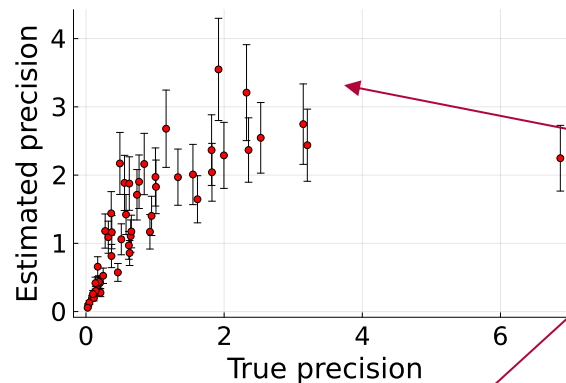
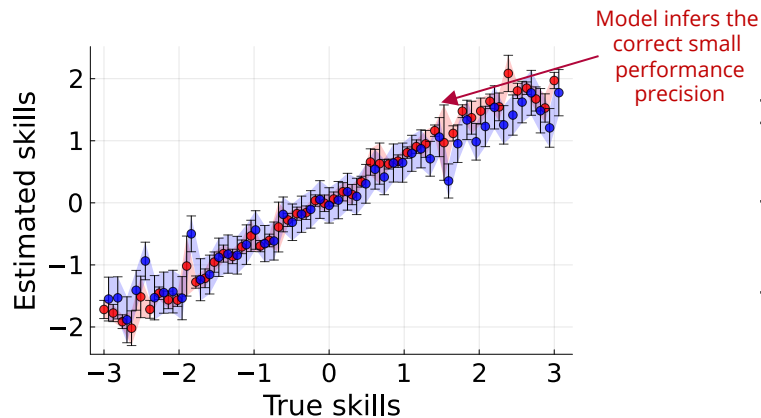


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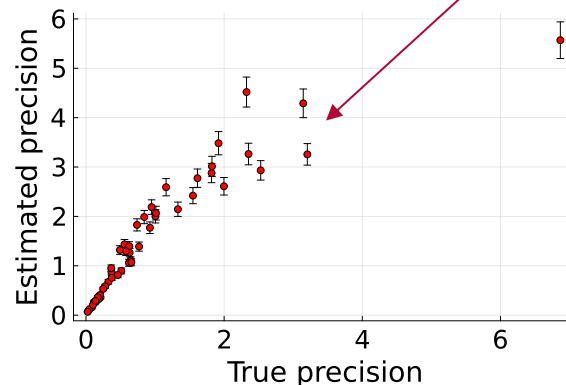
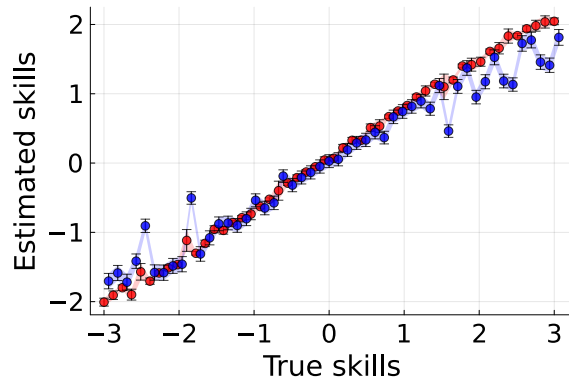
*Unit 6 – Approximate Inference:
Variational Inference*

Robust Bayesian Ranking: Per-Player Precision Example

$G = 10,000$



$G = 100,000$



Advanced Probabilistic Machine Learning

*Unit 6 – Approximate Inference:
Variational Inference*

■ Minimizing Reverse KL-Divergence via ELBO

- Minimizing the KL-divergence via expected sufficient statistic matching on the marginal can generate invalid message
- One alternative is to minimize the *reverse* KL divergence but there is no closed-form solution for the minimizer
- The solution is obtained by iterating fixed-point equations until convergence

■ Variational Inference for Gaussian Factors

- One of the most prominent examples where variational message passing works if the Gaussian density when the precision (i.e., inverse variance) has a Gamma distribution
- Variational message passing leads to an underestimate of the variance!

■ Robust Bayesian Regression and Ranking

- Bayesian regression with Gamma-distributed precision *per observation* becomes robust to outliers
- TrueSkill can also be extended with variable performance precision *per match* or *per player*. These algorithms are more robust (and have not yet been published!)

See you next week!